535510 Reinforcement Learning

Pre-Lecture Assignment 2

Instructor: Ping-Chun Hsieh
Email: pinghsieh@nycu.edu.tw

Department of CS, NYCU

Global Convergence of Exact PG and Stochastic PG

In Lectures 6-8, we will learn the stochastic PG algorithms and the global convergence of exact PG method. You can refer to the paper https://arxiv.org/abs/2005.06392 by Mei et al. in ICML 2020 and the slides of Lecture 6 to answer the following questions.

1) Non-uniform Polyak-Łojasiewicz (PL): To prove the convergence rate of exact PG, (Mei et al., 2020) established the non-uniform PL condition in Lemma 8 (also shown below).

(a) Can you explain why this is called "*non-uniform*"?

$$\left\| \frac{\partial V^{\pi_\theta}(\mu)}{\partial \theta} \right\|_2 \geq \frac{\min_s \pi_\theta(a^*(s)|s)}{\sqrt{S} \cdot \left\| d_\rho^{\pi^*} / d_\mu^{\pi_\theta} \right\|_\infty} \cdot [V^*(\rho) - V^{\pi_\theta}(\rho)]$$

(b) Where does the term $\min_{s \in \mathcal{S}} \pi_\theta(a^*(s) \mid s)$ come from? (Hint: Check the proof of Lemma 8)

(c) What's the difference between the distributions μ and ρ?

2) Q-value actor-critic (QAC) algorithm: The **QAC.py** is a Python file generated by GPT that implements the QAC method (see Lecture 6 slides).

(a) Can you convert Lines 100-119 into math equations and briefly explain your thought?

(b) Regarding the design in Line 112, do you agree with GPT?

Submission Guidelines and Remarks

- Please write all your responses within **1 page** (**A4 paper**, either hand-written or typeset). The responses that go beyond 1 page will *not* be graded.

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.

- Please submit it by **9pm, 3/24 (Tuesday)**.